

Customer Churn Prediction Model Technical Whitepaper

Model Architecture, Processing Pipeline and Business Integration

Prepared by: Richard McNerney
Date: March 2026

1. Overview

This document provides a detailed technical overview of the machine learning model developed to predict customer churn within a telecommunications dataset.

- Data ingestion and preprocessing
- Feature engineering and encoding
- Model training and evaluation
- Prediction generation
- Business-ready output creation

2. Model Pipeline Summary

Data Ingestion → Data Cleaning → Feature Encoding → Train/Test Split → Scaling → Model Training → Evaluation → Prediction → Business Output

3. Detailed Cell-by-Cell Breakdown

Cell 1 — Data Acquisition

Downloads dataset via KaggleHub to ensure reproducibility and remove manual dependencies.

Cell 2 — Data Loading & Exploration

Loads dataset into pandas, validates structure, checks missing values.

Cell 3 — Data Cleaning

Converts data types, handles missing values, removes ID fields to prevent leakage.

Cell 4 — Feature Encoding

Applies one-hot encoding and binary target transformation.

Cell 5 — Train/Test Split

Splits dataset into training/testing sets using stratification.

Cell 6 — Feature Scaling

Applies StandardScaler for normalization.

Cell 7 — Model Training

Trains multiple models and selects best performer.

Cell 8 — Feature Importance

Extracts and visualizes key drivers of churn.

Cell 9 — Model Evaluation

Evaluates model using confusion matrix.

Cell 10 — Model Persistence

Saves trained model using joblib.

Cell 11 — Predictions

Generates churn predictions and probabilities.

Cell 12 — Business Logic

Creates risk bands and retention priorities.

Cell 13 — Export

Exports predictions to CSV for BI integration.

Cell 15 — Top Customers

Identifies top 100 high-risk customers.

Cell 17 — Feature Importance Export

Exports feature importance for dashboard.

4. Model Design Considerations

- Random Forest selected for performance and interpretability
- Feature engineering ensures numeric consistency
- Business translation layer enables usability

5. Integration with Power BI

- Predictions feed KPIs and segmentation
- Top 100 customers support retention actions
- Feature importance drives ML insights

6. Strengths

- End-to-end pipeline
- Interpretable outputs
- Business-ready insights
- Scalable architecture

7. Limitations

- Static dataset
- Limited tuning
- External factors excluded

8. Future Enhancements

- Hyperparameter tuning
- Time-series modelling
- CLV integration
- Automated retraining
- Real-time scoring

9. Conclusion

This model enables proactive churn prediction and supports data-driven retention strategies aligned with business goals.